

Indian Institute of Technology (IIT-Bombay)

AUTUMN Semester, 2026

COMPUTER SCIENCE AND ENGINEERING

CS230: Digital Logic Design + Computer Architecture

Quiz I

Full Marks: 15

Time allowed: 0.75 hours

1. The truth table for a half adder is given below:

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Construct a full adder using two half adders and one extra gate. Draw the diagram. (5 marks)

Solution:

Grading Rule: Give marks if the diagram is fully correct. If someone gives correct expression (in terms of half adders) but no diagram, give 4. Else give 0.

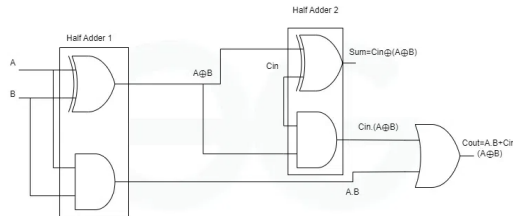


Figure 1: Full adder with extra AND gate

2. **Tricky:** Let

$$f = (f_1 \cdot f_2) + f_3,$$

where

$$f(w, x, y, z) = \sum m(0, 4, 9, 10, 11, 12)$$

and

$$f_1 = x'z + xz'.$$

Determine the Boolean functions f_2 and f_3 subject to the following conditions:

1. $f_1.f_2$ must cover all minterms of f for which $f_1 = 1$.
2. f_3 must cover exactly the remaining minterms of f that are not covered by f_1f_2 .
3. There must be no overlap between the minterms covered by f_1f_2 and f_3 .
4. Express both f_2 and f_3 in minimum Sum-of Products (SOP) form.

(5 marks)

Solution: We are given

$$f = (f_1 \wedge f_2) \vee f_3,$$

where

$$f(w, x, y, z) = \sum m(0, 4, 9, 10, 11, 12)$$

and

$$f_1 = x'z + xz'.$$

First, note that

$$f_1 = x'z + xz' = x \oplus z.$$

Hence, $f_1 = 1$ whenever $x \neq z$.

Now consider the minterms for which $f = 1$:

Minterm	w	x	y	z	f_1
0	0	0	0	0	0
4	0	1	0	0	1
9	1	0	0	1	1
10	1	0	1	0	0
11	1	0	1	1	1
12	1	1	0	0	1

Therefore, f_1f_2 must cover the minterms

$$4, 9, 11, 12.$$

To determine f_2 , the input combinations for which $f_1 = 0$ may be treated as don't-care conditions, since the value of f_2 does not affect f_1f_2 at those points. Minimizing with these don't-care conditions gives

$$f_2 = wx' + xy',$$

$$f_2 = wx' + y'z',$$

$$f_2 = wz + xy',$$

$$f_2 = wz + y'z'.$$

Thus, f_1f_2 covers exactly the minterms

$$4, 9, 11, 12.$$

The remaining minterms of f are

$$0, 10.$$

Since f_3 must cover exactly these minterms and there can be no overlap with f_1f_2 ,

$$f_3 = m_0 + m_{10}.$$

Now,

$$m_0 = w'x'y'z'$$

and

$$m_{10} = wx'yz'.$$

Hence,

$$f_3 = w'x'y'z' + wx'yz'.$$

Therefore, the required Boolean functions are

$$f_2 = wx' + xy',$$

$$f_2 = wx' + y'z',$$

$$f_2 = wz + xy',$$

$$f_2 = wz + y'z'.$$

and

$$f_3 = w'x'y'z' + wx'yz'.$$

Grading Rule: Please give 3 marks if any one the f_2 s has been found correctly. Give 2 marks if the don't cares are not considered but the main minterms of f_2 have been identified and showed as equations f_1f_2 . Give 2 marks for f_3 . Give 4 marks if some other correct solution is found.

3. A 4-bit synchronous binary counter is driven by a 16 MHz clock. The states are Q_0 , Q_1 , Q_2 , Q_3 , where Q_0 is the least significant bit (LSB). You are asked to derive a 4 MHz clock from this counter. How would you do it? (5 marks)

Solution: A 4-bit synchronous binary counter is driven by a clock of frequency

$$f_{\text{clk}} = 16 \text{ MHz}.$$

Let the counter outputs be $Q_3Q_2Q_1Q_0$, where Q_0 is the least significant bit (LSB). The counting sequence of the counter is shown below:

Count	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

From the table, the output sequence at Q_1 is

$$Q_1 : 0, 0, 1, 1, 0, 0, 1, 1, \dots$$

Thus, Q_1 remains LOW for two input clock cycles and HIGH for two input clock cycles. Therefore, one complete cycle of Q_1 requires four cycles of the input clock. Hence,

$$f_{Q_1} = \frac{f_{\text{clk}}}{4}.$$

Since

$$f_{\text{clk}} = 16 \text{ MHz},$$

we obtain

$$f_{Q_1} = \frac{16 \text{ MHz}}{4} = 4 \text{ MHz}.$$

Therefore, the required 4 MHz clock can be obtained directly from the Q_1 output, i.e., the second least significant bit of the counter:

$$\boxed{f_{\text{out}} = f_{Q_1} = 4 \text{ MHz}}.$$

Grading Rule: Give 4 marks if someone gets the frequency division idea. Give 5 marks if someone also specifies the bit from which the clock should be derived. No need for proper circuit etc.
